

Quantum Computing Research Portfolio

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Quantum Information Theory, Complexity, and AI

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github.com/asmeeta-quantum/ETH_Qiskit_Workspace

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Abstract

This portfolio synthesizes my independent research across three interconnected domains: (1) quantum foundations and information theory, (2) quantum complexity and verification, and (3) quantum computing and artificial intelligence. My work combines analytical modeling of open quantum systems with numerical simulations using Qiskit and QuTiP, alongside exploration of AI-driven methods for quantum algorithm discovery and error mitigation. Background at ISRO provides practical insight into physical noise sources in cryogenic and microwave systems relevant for quantum devices.

Research Goal: Develop rigorous frameworks for understanding the fundamental limits of noisy quantum computation, with applications in verification, error mitigation, and quantum-enhanced AI.

Research Domains

Domain 1: Quantum Foundations and Information Theory

- Quantum information theory and open quantum systems
- Noise characterization and quantum channel divergences
- Verification complexity under realistic noise
- T_1/T_2 relaxation and decoherence analysis

Domain 2: Quantum Complexity and Verification

- Complexity-theoretic analysis of quantum advantage
- Classical simulability boundaries under noise
- Verification protocols and sample complexity
- Resource theories for noisy quantum computation

Domain 3: Quantum Computing and Artificial Intelligence

- AI-driven quantum error mitigation and optimization
- Hybrid quantum-classical machine learning
- Quantum-enhanced AI for combinatorial optimization
- Neural networks for noise characterization

Key Research Contributions

Verification Complexity Under Noise

Objective: Understand how T_1/T_2 relaxation and dephasing affect verification sample complexity.

Methods:

- Analytical scaling bounds for verification under noise channels
- Quantum channel divergences and entropy measures
- Identification of noise thresholds for classical simulability

Key Result: Derived analytical bounds identifying regimes where quantum advantage becomes classically simulable.

Error Mitigation Theory and Simulation

Objective: Develop and validate error mitigation frameworks for noisy quantum circuits.

Methods:

- Zero-Noise Extrapolation (ZNE) implementation
- Systematic noise amplification analysis
- Benchmarking on simulated noisy circuits

Key Result: Demonstrated $\approx 20\%$ fidelity improvement on noisy quantum circuits.

AI-Informed Quantum Control Optimization

Objective: Suppress non-computational excitations in weakly anharmonic transmon qubits using AI-driven optimization.

Methods:

- DRAG pulse optimization framework
- Lindblad master equation simulations
- Leakage-aware control theory

Key Result: Achieved $> 90\%$ suppression of $|1\rangle \rightarrow |2\rangle$ leakage.

Hardware-Informed Noise Modeling

Objective: Translate microwave component stability data into quantum noise parameters.

Methods:

- Phase noise characterization from VNA measurements
- Cryogenic system stability analysis (4K)
- Composite noise channel modeling

Key Insight: Hardware characterization data provides practical constraints for quantum noise models, bridging theory and experiment.

AI-Assisted Quantum Error Mitigation

Objective: Use neural networks to optimize quantum error mitigation parameters.

Methods:

- MLPRegressor for ZNE parameter optimization
- Feature importance analysis for noise characterization
- Synthetic data generation for training

Key Result: Neural network-driven optimization for improved quantum circuit fidelity.

Hybrid Quantum-Classical Classification

Objective: Demonstrate hybrid approaches where quantum circuits serve as feature extractors for classical ML.

Methods:

- Quantum kernel simulation
- Random Forest classifier baseline
- Hybrid accuracy comparison

Key Result: Hybrid quantum-classical models show potential for improvement over purely classical approaches.

Research Portfolio Structure

ETH_Qiskit_Workspace/

1. 01_bell_states_entanglement.ipynb
2. 02_statevector_simulation.ipynb
3. 03_depolarizing_noise_analysis.ipynb
4. 04_superconducting_noise_channels.ipynb (T_1/T_2 modeling)
5. 05_composite_noise_simulation.ipynb
6. 06_grover_verification.ipynb
7. 07_deutsch_jozsa_advantage.ipynb
8. 08_verification_sample_complexity.ipynb (Key: analytical bounds)
9. 09_lindblad_qubit_dynamics.ipynb
10. 10_drag_pulse_optimization.ipynb (Key: $> 90\%$ suppression)
11. 11_noise_bench_qc/ (Benchmarking suite)
12. 12_lindblad_sim/ (Simulation toolkit)
13. 13_ai_error_mitigation/ (AI-driven ZNE)
14. 14_hybrid_qml/ (Hybrid quantum ML)
15. 15_qaoo/ (QAOA optimization)

Technical Implementation

Software Tools Developed

Tool	Description	Key Result
Noise-Bench QC	Benchmarking suite for verification under realistic noise	Systematic noise characterization framework
Lindblad-Sim	Non-Markovian and leakage-aware simulation toolkit	Open system dynamics simulation
ZNE Toolkit	Zero-Noise Extrapolation framework	$\approx 20\%$ fidelity improvement
DRAG Pulse Optimizer	Adaptive pulse design for leakage suppression	$> 90\%$ leakage suppression
AI-Error Mitigation	Neural network-driven error mitigation optimization	Automated ZNE parameter tuning
Hybrid-QML	Hybrid quantum-classical ML framework	Quantum-enhanced feature extraction

Technical Stack

Quantum Frameworks	Qiskit (Terra, Aer, Pulse), QuTiP, OpenQASM
Programming	Python (NumPy, SciPy, Matplotlib, scikit-learn)
AI/ML	TensorFlow, PyTorch, Scikit-Learn
Quantum Theory	Lindbladian models, smooth entropies, quantum channel divergences, resource theories
Mathematics	Linear algebra, probability, information theory, optimization

Hardware Relevance

Professional experience at ISRO with:

- Cryogenic systems (4K) and microwave component characterization
- VNA S-parameter analysis and stability metrics (T_1 , T_2 , phase noise)
- Low-noise amplifier characterization for quantum readout systems

Key Achievements

1. Analytical bounds linking physical noise (T_1/T_2) to verification scalability
2. Open-source tools for realistic quantum noise simulation
3. Hardware-to-quantum modeling pipeline from ISRO experience
4. Identification of noise-triggered simulability transitions
5. $> 90\%$ leakage suppression via DRAG pulse optimization
6. $\approx 20\%$ fidelity improvement via zero-noise extrapolation
7. AI-driven frameworks for quantum error mitigation
8. Hybrid quantum-classical ML models for optimization

Research Alignment with ETH Zürich

This work supports PhD applications to ETH Zürich across three positions:

Position	Supervisor(s)	Research Area
Dominik Hangleiter	Prof. Dominik Hangleiter	Quantum complexity, verification, noisy quantum systems
ETH + IBM Research	Dr. David Sutter, Prof. Renato Renner, Dr. Elisa Bäumer, Dr. Christophe Piveteau	Quantum information theory, error mitigation, foundations
AI × Quantum Computing	Dr. Stefan Woerner, Prof. Juan Carrasquilla	AI-driven quantum algorithms, optimization, hybrid models

Focus Areas:

- Verification complexity of quantum computations under realistic noise

- Practical quantum advantage certification in NISQ devices
- Resource theories for noisy quantum computation
- Classical simulability thresholds and noise-adapted advantage
- AI-driven quantum error mitigation and optimization

Repository and Code Availability

Full Repository: https://github.com/asmeeta-quantum/ETH_Qiskit_Workspace

All notebooks, simulation toolkits, benchmarking suites, and AI/ML frameworks are available for review.